

S HARI PREETHAM

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SUMMARY

Data-Driven Mechanical Engineer specializing in systems engineering for robotics and control systems. Experienced in developing sensor fusion and control algorithms, 3D localization systems, and Design for bridging mechanical domain knowledge with software engineering for intelligent physical systems.

EDUCATION

University College of Engineering, Science & Technology – JNTUH Hyderabad
Integrated Dual Degree in Mechanical Engineering (Advanced Manufacturing Systems) Aug 2021 – Apr 2026
GPA: 8/10

Narayana Junior College Hyderabad
Intermediate (Mathematics, Physics, Chemistry) Apr 2019 – Apr 2021
Score: 97.2%

EXPERIENCE

Indian Institute of Technology, Madras Chennai
Project Associate Feb 2026 – Present
Guides: Dr. S. Ganga Prasath and Dr. Danny Raj Masila
FedEx-Sponsored Research Project: Bio-Inspired Navigation for Autonomous Warehouse Robots

- Researching the statistical physics of Reinforcement Learning by modelling entropy-regularised policy optimisation as a stochastic optimal control process, with emphasis on cost-landscape evolution, transport dynamics, and interpretability of emergent navigation policies.
- Building high-performance RL training pipelines in JAX using `jit` compilation and `vmap` for large-scale vectorised environment rollouts.
- Developed interactive visualisation tools for RL interpretability, including trajectory rendering, policy-switching heatmaps, and transfer-matrix analysis.

Defence Research and Development Organisation, DRDO-DRDL Hyderabad
Project Intern Sept 2023 – Oct 2023

- Developed Python-based NumPy/SciPy pipelines to analyze 13,000+ data points for MIRV aerodynamic heating, significantly improving computational efficiency over legacy MATLAB workflows.
- Implemented PyROOT for advanced processing of isentropic flow properties and stagnation temperatures to enable high-fidelity parametric studies of transient heat behavior.
- Provided data-backed insights into thermal protection and trajectory design, supporting the validation of critical re-entry vehicle components under Scientist ‘F’ Shaik Ismail.

Indian Institute of Technology, BHU Varanasi
Workshop Trainee June 2023

- Performed structural and thermal simulations in ANSYS and CAD modeling for component optimization and reverse engineering.
- Applied topology optimization and Design-for-Additive-Manufacturing (DFAM) across FDM, SLA, and SLS processes to prototype lightweight, high-strength structures.

RESEARCH EXPERIENCE

Real-Time Vibration Compensation in Low-Cost Laser Steering Systems via Visual-Inertial Fusion Jun 2025 – Present
Supervisor: Dr. A. Chennakesava Reddy
[Github Repository](#)

- Project aims to develop a line-of-sight stabilization algorithm for laser beam steering applications (e.g., UCAS, precision agriculture laser weeding).
- Designed and fabricated a 2-axis galvo scanner using salvaged Voice Coil Motors (VCMs); CAD modeling and open-loop operation demonstrated.

- Developed a simulation environment (PyQtGraph + OpenGL) to model kinematics, ray-plane intersection, and laser scanning control.
- Built real-time object detection and tracking pipeline using YOLOv8 + DeepSORT for visual servoing.
- Implemented sensor fusion (IMU + camera) and control algorithms (PID, LQR) for vibration rejection and precision pointing.

PROJECTS

Bio-Inspired Navigation Algorithm for Featureless Environments [Github](#)

- Developed a simulation framework to address perceptual aliasing in warehouses where Visual SLAM fails due to lack of unique landmarks.
- Implemented a bio-mimetic spectral compass (Inspired from Cataglyphis ants) to correct stochastic sensor drift using sparse global cues, enabling infrastructure-free navigation.
- Implemented a state estimator (EKF) to fuse odometry with a spectral compass, Modeled sensor noise using Gaussian distributions, tracking the system's internal state by propagating a 3x3 covariance matrix (P).
- Achieved an 81% reduction in terminal positioning error compared to blind dead-reckoning by decoupling orientation from translation.

Real-Time Multi-Camera Motion Capture and 3D Localization System [Github](#)

- Developed an indoor motion capture system utilizing calibrated high-speed PS3 Eye cameras (operating at 187 fps) to capture motion data.
- Implemented principles of epipolar geometry for accurate and real-time 3D localization of physical markers within the capture volume.
- Applied system identification techniques on captured motion data to characterize dynamic behavior.

ROS2 Visual Processor for Real-Time Robotics [Github](#)

- Developed a low-latency visual perception system in ROS2 for control and vision for remote-operated robotics.
- Integrated YOLOv8 on a ROS2 node using WebRTC for real-time object detection visualization.
- Optimized system for deterministic performance using the Linux PREEMPT_RT patch.

Quadruped Spider Robot [Github](#)

- Designed and built a Bluetooth-controlled quadruped robot using an Arduino Uno and 8 servo motors.
- Implemented inverse kinematics for precise leg positioning and developed multiple gait generation algorithms for stable locomotion.
- Programmed kinematic calculations and control loops for seamless operation via a custom mobile application.

TECHNICAL SKILLS

Programming:	Python, C++
Robotics Systems:	ROS2, Sensor Fusion, State Estimation (EKF, UKF)
Machine Learning:	Reinforcement Learning, Deep Learning, YOLOv8, DeepSORT
Computer Vision:	OpenCV, Multi-View Geometry, Object Tracking
Simulation & Modeling:	OpenGL, Gazebo, ANSYS, Fusion 360
Scientific Computing:	NumPy, SciPy
Tools:	Linux, Git, Docker